ARJUNA JEE 2027

Lecture -12 | Mathematics

TRIGONOMETRIC FUNCTIONS

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Topics *to be covered*

A **Trigonometric Series**

B **Graph of Trigonometric Functions**

C **Question Practice**

D



H.W. Discussion

Let $S = \sum_{r=1}^{90} (2r)\sin(2r)^\circ$ then value of $S/90$ is (are) :

A $\cot 1^\circ$

B $\tan 1^\circ$

C $\dfrac{\cos^2 20^\circ - \sin^2 19^\circ}{\sin^2 26^\circ - \sin^2 25}$

D $\sqrt{\dfrac{1-\cos 2^\circ}{1+\cos 2^\circ}}$

$S = 2\sin 2^\circ + 4\sin 4^\circ + 6\sin 6^\circ + \ldots\ldots\ldots + 176\sin 176^\circ + 178\sin 178^\circ + 180\sin 180^\circ$

$S = 180\left(\sin 2^\circ + \sin 4^\circ + \sin 6^\circ + \ldots\ldots + \sin 88^\circ\right) + 90$

$\alpha = 2^\circ,\ \beta = 2^\circ,\ \eta = 44 \rightarrow \dfrac{\beta}{2} = 1^\circ$

$180\,\dfrac{\sin(44\times 1^\circ)\sin(2^\circ + 43\times 1^\circ)}{\sin 1^\circ} + 90$

$180\,\dfrac{\sin 44^\circ \sin 45^\circ}{\sin 1^\circ} + 90$

$$\frac{180}{2}\left(\frac{2\sin 44^\circ \sin 45^\circ}{\sin 1^\circ}\right)+90$$

$$90\left(\frac{\cos(1^\circ)-\cos(89^\circ)}{\sin 1^\circ}\right)+90.$$

$$90\left(\cot 1^\circ - 1\right)+90$$

$$S = 90\cot 1^\circ - 90 + 90$$

$$\frac{S}{90} = \cot 1^\circ$$

$$S = 2\sin 2^\circ + 4\sin 4^\circ + 6\sin 6^\circ + \ldots\ldots + 176\sin 176^\circ + 178\sin 178^\circ + 180\sin 180^\circ$$

$r=44$ $r=45$ $r=46$

$$88\sin 88^\circ + \boxed{90\sin(90^\circ)} + 92\sin 92^\circ$$

$\sin 92^\circ = \sin 88^\circ$

$$180\sin 88^\circ.$$

Let $S = \sum_{r=1}^{90} (2r)\sin(2r)^\circ$ then value of $S/90$ is (are) :

(A) cot 1°

(B) tan 1°

(C) $\dfrac{\cos^2 20^\circ - \sin^2 19^\circ}{\sin^2 26^\circ - \sin^2 25}$

(D) $\sqrt{\dfrac{1-\cos 2^\circ}{1+\cos 2^\circ}}$

(C) $\dfrac{\cos(20^\circ+19^\circ)\cdot\cos(20^\circ-19^\circ)}{\sin(26^\circ+25^\circ)\cdot\sin(26^\circ-25^\circ)} = \dfrac{\cos 39^\circ \cdot \cos 1^\circ}{\sin 51^\circ \sin 1^\circ}$

(D) $\sqrt{\dfrac{1-\cos 2^\circ}{1+\cos 2^\circ}} = \sqrt{\dfrac{2\sin^2 1^\circ}{2\cos^2 1^\circ}} = \dfrac{|\sin 1^\circ|}{|\cos 1^\circ|} = \tan 1^\circ$

TRIGONOMETRIC SERIES

Type-02 : Splitting T_n as difference of two terms (Viral + Trending)

(1) $$\frac{1}{\sin A \cdot \sin B} \times \frac{\sin(A-B)}{\sin(A-B)} = \left(\frac{\sin A \cos B}{\sin A \sin B} - \frac{\cos A \sin B}{\sin A \sin B}\right) \times \frac{1}{\sin(A-B)}$$

$$= \frac{1}{\sin(A-B)} (\cot B - \cot A)$$

(2) $$\frac{1}{\cos A \cdot \cos B} \times \frac{\sin(A-B)}{\sin(A-B)} = \frac{1}{\sin(A-B)} \times \left(\frac{\sin A \cos B}{\cos A \cos B} - \frac{\cos A \sin B}{\cos A \cos B}\right)$$

$$= \frac{1}{\sin(A-B)} (\tan A - \tan B)$$

(1) $\dfrac{1}{\sin A \cdot \sin B} = \dfrac{1}{\sin(A-B)} (\cot B - \cot A)$

(2) $\dfrac{1}{\cos A \cdot \cos B} = \dfrac{1}{\sin(A-B)} (\tan A - \tan B)$

ex:- $\dfrac{1}{\sin \underbrace{13^\circ}_{B} \sin \underbrace{17^\circ}_{A}} = \dfrac{1}{\sin(4^\circ)} (\cot 13^\circ - \cot 17^\circ)$

ex:- $\dfrac{1}{\cos \underbrace{13^\circ}_{B} \cos \underbrace{14^\circ}_{A}} = \dfrac{1}{\sin(1^\circ)} \{\tan 14^\circ - \tan 13^\circ\}$


Blue बादल
Hint: Trigonometric Terms are in multiplication in Denominator
What to do??
multiply & divide each Term by sin(difference b/w angles)

Question

The sum of the series $\sum_{r=0}^{88} \sec r^\circ \cdot \sec(r+1)^\circ$ is equal to

A $\operatorname{cosec} 1^\circ \cdot \tan 1^\circ$

B $\operatorname{cosec} 1^\circ \cdot \cot 1^\circ$

C $\sec 1^\circ \cdot \tan 1^\circ$

D $\sec 1^\circ \cdot \cot 1^\circ$

$$\# \quad \frac{1}{\cos r^\circ \cos(r+1)^\circ} = \frac{1}{\sin 1^\circ}\left(\tan(r+1)^\circ - \tan r^\circ\right)$$

$r=0 \Rightarrow T_1 = \frac{1}{\sin 1^\circ}\left(\tan 1^\circ - \tan 0^\circ\right)$

$r=1 \Rightarrow T_2 = \frac{1}{\sin 1^\circ}\left(\tan 2^\circ - \tan 1^\circ\right)$

$r=2 \Rightarrow T_3 = \frac{1}{\sin 1^\circ}\left(\tan 3^\circ - \tan 2^\circ\right)$

$\vdots$

$r=88 \Rightarrow T_{89} = \frac{1}{\sin 1^\circ}\left(\tan 89^\circ - \tan 88^\circ\right)$

add $\Rightarrow \frac{1}{\sin 1^\circ}\left(\tan 89^\circ - 0\right)$

$= \operatorname{cosec} 1^\circ \cdot \cot 1^\circ$

Question [JEE (Adv.)-2016]

The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4}+\frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4}+\frac{k\pi}{6}\right)}$ is equal to

A $3-\sqrt{3}$

B $2(3-\sqrt{3})$

C $2(\sqrt{3}-1)$

D $2(2+\sqrt{3})$

$B = \frac{\pi}{4} + \frac{k\pi}{6} - \frac{\pi}{6}$

$B = A - \frac{\pi}{6}$

$30^\circ = \frac{\pi}{6} = A - B$

$\frac{1}{2} = \sin(A-B)$

$\# \quad \frac{1}{\sin(A-B)} = 2.$

$A.$

$\frac{1}{\sin B \sin A} = \frac{2}{\sin(A-B)} \{\cot B - \cot A\}$

$\sum_{k=1}^{13} 2\left\{\cot\left(\frac{\pi}{4}+(k-1)\frac{\pi}{6}\right) - \cot\left(\frac{\pi}{4}+\frac{k\pi}{6}\right)\right\}$

$$\sum_{k=1}^{13} 2\left\{\cot\left(\frac{\pi}{4}+(k-1)\frac{\pi}{6}\right)-\cot\left(\frac{\pi}{4}+\frac{k\pi}{6}\right)\right\}$$

$k=1 \Rightarrow T_1 = 2\left(\cot\frac{\pi}{4} - \cot\left(\frac{\pi}{4}+\frac{\pi}{6}\right)\right)$

$T_2 = 2\left(\cot\left(\frac{\pi}{4}+\frac{\pi}{6}\right) - \cot\left(\frac{\pi}{4}+\frac{2\pi}{6}\right)\right)$

$T_3 = 2\left(\cot\left(\frac{\pi}{4}+\frac{2\pi}{6}\right) - \cot\left(\frac{\pi}{4}+\frac{3\pi}{6}\right)\right)$

$\vdots$

$T_{13} = 2\left(\cot\left(\frac{\pi}{4}+12\frac{\pi}{6}\right) - \cot\left(\frac{\pi}{4}+\frac{13\pi}{6}\right)\right)$

add $\Rightarrow 2\left(1 - \cot\frac{5\pi}{12}\right)$

$\cot\left(\frac{\pi}{4}+\frac{13\pi}{6}\right)$

$= \cot\left(\frac{\pi}{4}+2\pi+\frac{\pi}{6}\right)$

$= \cot\left(2\pi+\frac{5\pi}{12}\right)$

$= \cot\frac{5\pi}{12}$

Question [JEE (Adv.)-2016]

The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4}+\frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4}+\frac{k\pi}{6}\right)}$ is equal to

A $3-\sqrt{3}$

B $2(3-\sqrt{3})$

C $2(\sqrt{3}-1)$

D $2(2+\sqrt{3})$

$$2\left(1-\cot\frac{5\pi}{12}\right) \quad \left(\frac{5\pi}{12} = 15^\circ\right)$$

$$2\left(1-\cot 75^\circ\right)$$

$$2\left(1-\tan 15^\circ\right)$$

$$2\left(1-\left(2-\sqrt{3}\right)\right) = 2\left(\sqrt{3}-2+1\right)$$

$$= 2\left(\sqrt{3}-1\right)$$

Question [IIT-JEE-2009]

For $0 < \theta < \frac{\pi}{2}$, then solution(s) of

$$\sum_{m=1}^{6} \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right) \operatorname{cosec}\left(\theta + \left(\frac{m\pi}{4}\right)\right) = 4\sqrt{2}$$ is(are)

A $\frac{\pi}{4}$

B $\frac{\pi}{6}$

C $\frac{\pi}{12}$

D $\frac{5\pi}{12}$

Say: $A = \theta + \frac{(m-1)\pi}{4}$ & $B = \theta + \frac{m\pi}{4}$

$A = \theta + \frac{(m\pi - \pi)}{4}$

$A = \theta + \frac{m\pi}{4} - \frac{\pi}{4}$

$A = B - \frac{\pi}{4}$

$\frac{\pi}{4} = B - A$

$\operatorname{cosec}A \cdot \operatorname{cosec}B = \frac{1}{\sin A \cdot \sin B}$

$= \frac{\sin(B-A)}{\sin(B-A) \cdot \sin A \cdot \sin B}$

$= \frac{1}{\sin\frac{\pi}{4}} \left\{ \frac{\sin B \cos A}{\sin A \sin B} - \frac{\cos B \sin A}{\sin A \sin B} \right\}$

$= \sqrt{2}\left(\cot A - \cot B\right)$ ✓

$$\operatorname{cosec}A \cdot \operatorname{cosec}B = \sqrt{2}\left(\cot A - \cot B\right)$$

$$\sum_{m=1}^{6} \operatorname{cosec}A \cdot \operatorname{cosec}B = \sum_{m=1}^{6} \sqrt{2}\left\{\cot\left(\theta + (m-1)\frac{\pi}{4}\right) - \cot\left(\theta + \frac{m\pi}{4}\right)\right\} = 4\sqrt{2}$$

$$T_1 = \sqrt{2}\left(\cot\theta - \cot\left(\theta + \frac{\pi}{4}\right)\right)$$

$$T_2 = \sqrt{2}\left(\cot\left(\theta + \frac{\pi}{4}\right) - \cot\left(\theta + \frac{2\pi}{4}\right)\right)$$

$$T_3 = \sqrt{2}\left(\cot\left(\theta + \frac{2\pi}{4}\right) - \cot\left(\theta + \frac{3\pi}{4}\right)\right)$$

$$\vdots$$

$$T_6 = \sqrt{2}\left(\cot\left(\theta + \frac{5\pi}{4}\right) - \cot\left(\theta + \frac{6\pi}{4}\right)\right)$$

$\frac{6\pi}{4} \to \frac{3\pi}{2}$

$$\text{add} = \sqrt{2}\left(\cot\theta - \cot\left(\frac{3\pi}{2} + \theta\right)\right) = 4\sqrt{2}$$

$$\cot\left(\frac{3\pi}{2} + \theta\right) \downarrow$$

$$\left(\cot\theta - \left(-\tan\theta\right)\right) = 4$$

$$\cot\theta + \tan\theta = 4$$

$$\frac{1}{\tan\theta} + \tan\theta = 4$$

$$\frac{1 + \tan^2\theta}{\tan\theta} = 4.$$

$$\tan^2\theta - 4\tan\theta + 1 = 0.$$

$$\tan\theta = \frac{-(-4) \pm \sqrt{16 - 4}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$ ✓

Question [IIT-JEE-2009]

For $0 < \theta < \frac{\pi}{2}$, then solution(s) of

$$\sum_{m=1}^{6} \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right) \operatorname{cosec}\left(\theta + \left(\frac{m\pi}{4}\right)\right) = 4\sqrt{2}$$ is(are)

A $\frac{\pi}{4}$

B $\frac{\pi}{6}$

C $\frac{\pi}{12}$

D $\frac{5\pi}{12}$

$\tan\theta = 2-\sqrt{3}, 2+\sqrt{3}$

$\theta = \frac{\pi}{12}, \frac{5\pi}{12}$.

Challenger Question

Notes ✓

Prove that : $\dfrac{\sin x}{\cos 3x} + \dfrac{\sin 3x}{\cos 9x} + \dfrac{\sin 9x}{\cos 27x} = \dfrac{1}{2}(\tan 27x - \tan x)$

HHPP.

$$T_1 = \frac{2\sin x \cos x}{2\cos 3x \cos x} = \frac{1}{2}\frac{\sin 2x}{\cos 3x \cdot \cos x} \quad (\sin 2x = \sin(3x - x))$$

$$= \frac{1}{2}\left(\frac{\sin 3x \cos x}{\cos 3x \cos x} - \frac{\cos 3x \sin x}{\cos 3x \cos x}\right)$$

$$T_1 = \frac{1}{2}(\tan 3x - \tan x)$$

$$T_2 = \frac{2\sin 3x \cos 3x}{2\cos 9x \cdot \cos 3x} = \frac{1}{2}\frac{\sin 6x}{\cos 9x \cdot \cos 3x} \quad (\sin 6x = \sin(9x - 3x))$$

$$T_2 = \frac{1}{2}(\tan 9x - \tan 3x)$$

$$T_3 = \frac{1}{2}(\tan 27x - \tan 9x)$$

$$\frac{}{\frac{1}{2}(\tan 27x - \tan x)}$$

Critical Thinking Question (CTQ)

Prove that :

$$\frac{\cos 3x}{\sin 2x \sin 4x} + \frac{\cos 5x}{\sin 4x \sin 6x} + \frac{\cos 7x}{\sin 6x \sin 8x} + \frac{\cos 9x}{\sin 8x \sin 10x} = \frac{1}{2}(\operatorname{cosec} x)[\operatorname{cosec} 2x - \operatorname{cosec} 10x]$$



Question

Show that: $\sum_{r=1}^{n} \tan\left(\frac{x}{2^r}\right)\sec\left(\frac{x}{2^{r-1}}\right) = \tan x - \tan\frac{x}{2^n}$

HHPP.

$r=1 \Rightarrow T_1 = \tan\left(\frac{x}{2}\right)\cdot\sec x = \frac{\sin\frac{x}{2}}{\cos\frac{x}{2}\cos x} = \frac{\sin x\cdot\cos\frac{x}{2} - \cos x\cdot\sin\frac{x}{2}}{\cos\frac{x}{2}\cdot\cos x}$ ($x - \frac{x}{2}$)

$r=2 \Rightarrow T_2 = \tan\left(\frac{x}{4}\right)\sec\left(\frac{x}{2}\right) = \frac{\sin\frac{x}{4}}{\cos\frac{x}{4}\cdot\cos\frac{x}{2}}$ ($\frac{x}{2} - \frac{x}{4}$)

$T_1 = \tan\frac{x}{2^0} - \tan\frac{x}{2^1}$

$T_2 = \tan\frac{x}{2^1} - \tan\frac{x}{4}$ ($\frac{x}{4} = \frac{x}{2^2}$)

$T_3 = \tan\frac{x}{4} - \tan\frac{x}{8}$ ($\frac{x}{8} = \frac{x}{2^3}$)

$T_n = \tan\left(\frac{x}{2^{n-1}}\right) - \tan\left(\frac{x}{2^n}\right)$

$$\# \quad \operatorname{cosec}(\theta) = \frac{\sin\left(\theta - \frac{\theta}{2}\right)}{\sin\theta \cdot \sin\frac{\theta}{2}} = \frac{\sin\theta \cdot \cos\frac{\theta}{2}}{\sin\theta \cdot \sin\frac{\theta}{2}} - \frac{\cos\theta \cdot \sin\frac{\theta}{2}}{\sin\theta \cdot \sin\frac{\theta}{2}}$$

$$\boxed{\# \quad \operatorname{cosec}(\theta) = \cot\frac{\theta}{2} - \cot\theta}$$

IMPORTANT SERIES INITIATORS



(i) $\cot\theta - \tan\theta = 2\cot 2\theta$

(ii) $\cot(\theta/2) - \cot\theta = \operatorname{cosec}\theta$

RAR

$$\cot 2\theta = \frac{\cos 2\theta}{\sin 2\theta} = \frac{\cos^2\theta - \sin^2\theta}{2\sin\theta\cos\theta}$$

$$\# \quad \cot 2\theta = \frac{1}{2}(\cot\theta - \tan\theta)$$

OR

$$** \quad \boxed{\operatorname{cosec}\theta = \cot\frac{\theta}{2} - \cot\theta}$$

$$\cot\theta - 2\cot 2\theta = \tan\theta \quad (\times 2)$$

Question

Show that : $\sum_{r=0}^{n} \operatorname{cosec}(2^r x) = \cot\frac{x}{2} - \cot 2^n x$

$r=0 \Rightarrow T_1 = \operatorname{cosec} x = \cot\frac{x}{2} - \cot x$

$r=1 \Rightarrow T_2 = \operatorname{cosec} 2x = \cot x - \cot 2x$

$r=2 \Rightarrow T_3 = \operatorname{cosec} 4x = \cot 2x - \cot 4x$

$\vdots$

$r=n \Rightarrow T_{n+1} = \operatorname{cosec} 2^n x = \cot 2^{n-1} x - \cot 2^n x$

add

HHPP.

Acidic Question

Show that : $\sum_{r=0}^{n} \frac{1}{2^r}\tan\left(\frac{\theta}{2^r}\right) = \frac{1}{2^n}\cot\left(\frac{\theta}{2^n}\right) - 2\cot 2\theta$ ← HHP.

$r=0 \Rightarrow T_1 = 1 \cdot \tan\theta = \cot\theta - 2\cot 2\theta$ $\quad T_1 = \frac{1}{2^0}\cot\theta - 2\cot 2\theta$

$r=1 \Rightarrow T_2 = \frac{1}{2^1}\tan\frac{\theta}{2^1} = \frac{1}{2}\left(\cot\frac{\theta}{2} - 2\cot\theta\right)$ $\quad T_2 = \frac{1}{2^1}\cot\frac{\theta}{2} - \frac{1}{2^0}\cot\theta$

$T_3 = \frac{1}{2^2}\tan\frac{\theta}{2^2} = \frac{1}{2^2}\left(\cot\frac{\theta}{2^2} - 2\cot\frac{\theta}{2}\right)$ $\quad T_3 = \frac{1}{2^2}\cot\frac{\theta}{2^2} - \frac{1}{2^1}\cot\frac{\theta}{2}$

$\vdots$

$T_{n+1} = \frac{1}{2^n}\cot\frac{\theta}{2^n} - \frac{1}{2^{n-1}}\cot\frac{\theta}{2^{n-1}}$

Question

Prove that : $\sum_{r=0}^{n} \frac{\sin(3^r\theta)}{\cos(3^{r+1}\theta)} = \frac{\tan(3^{n+1}\theta)-\tan\theta}{2}$

Hew.
80% (DIBY)

Question

Compute the value of the sum $\sum_{r=1}^{n}\left(\frac{\tan(2^{r-1})}{\cos(2^r)}\right)$

A $\tan 2^n - 1$

B $\tan 2^n - \tan 1$

C $\tan 2^{n-1} - \tan 1$

D $\tan 2^n + \tan 1$

• Q is BY – (A)

Critical Thinking Question

[Ans: A, D]

Given $f(n,\theta)=\tan\frac{\theta}{2}(1+\sec\theta)(1+\sec 2\theta)\left(1+\sec 2^2\theta\right)\ldots.(1+\sec 2^n\theta)$ where $n\in\mathbb{N}$, then which of the following is correct?



A $f\left(2,\frac{\pi}{16}\right)=1$

B $f\left(2,\frac{\pi}{16}\right)=\sqrt{2}-1$

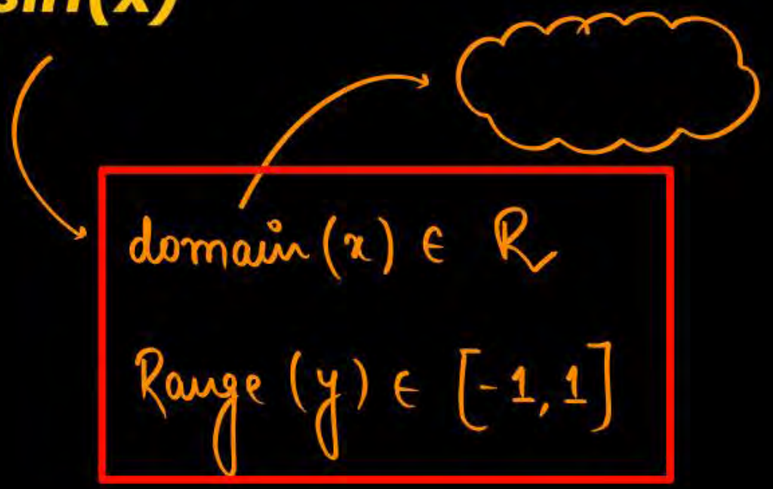
C $f\left(2,\frac{\pi}{32}\right)=1$

D $f\left(2,\frac{\pi}{32}\right)=\sqrt{2}-1$

GRAPHS OF TRIGONOMETRIC FUNCTIONS

Domain, Range & Graph of Six Trigonometric Functions

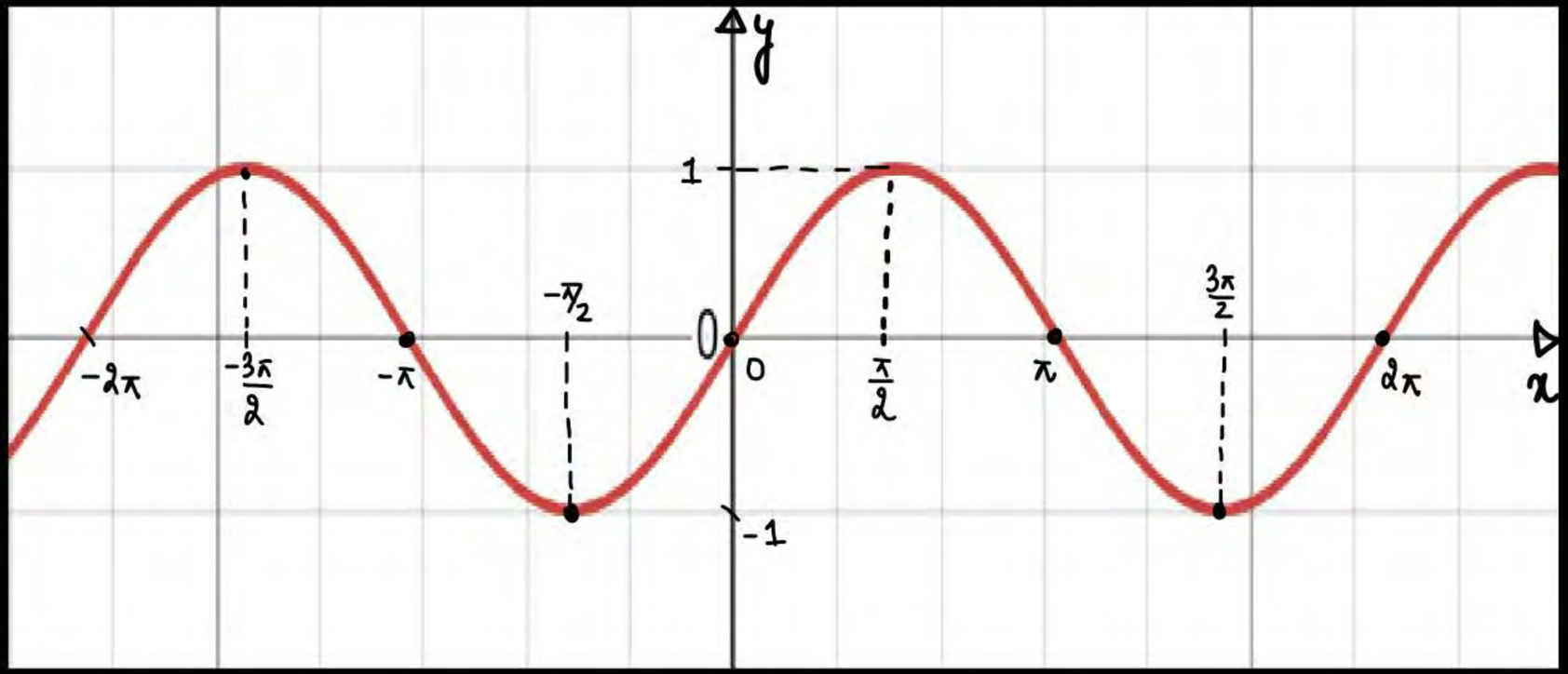
1. $y = \sin(x)$



GRAPHS OF TRIGONOMETRIC FUNCTIONS

Domain, Range & Graph of Six Trigonometric Functions

1. $y = \sin(x)$



Period = 2π.

RAPID FIRE ROUND

Application of graph:

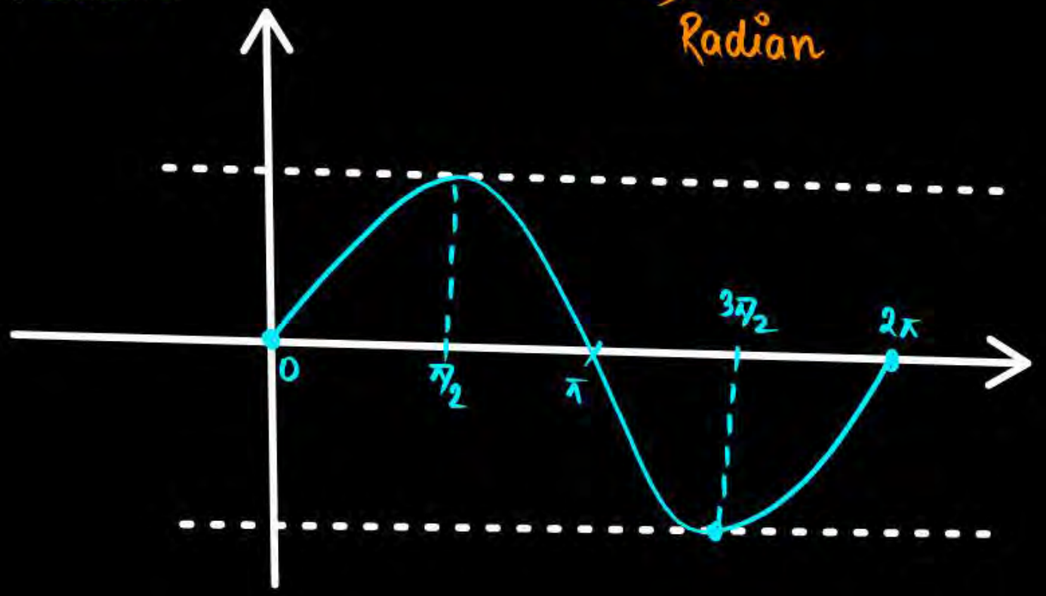
(i) For $x \in [0, 2\pi]$: $\sin x$ increases in $x \in \left[0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$ ✓

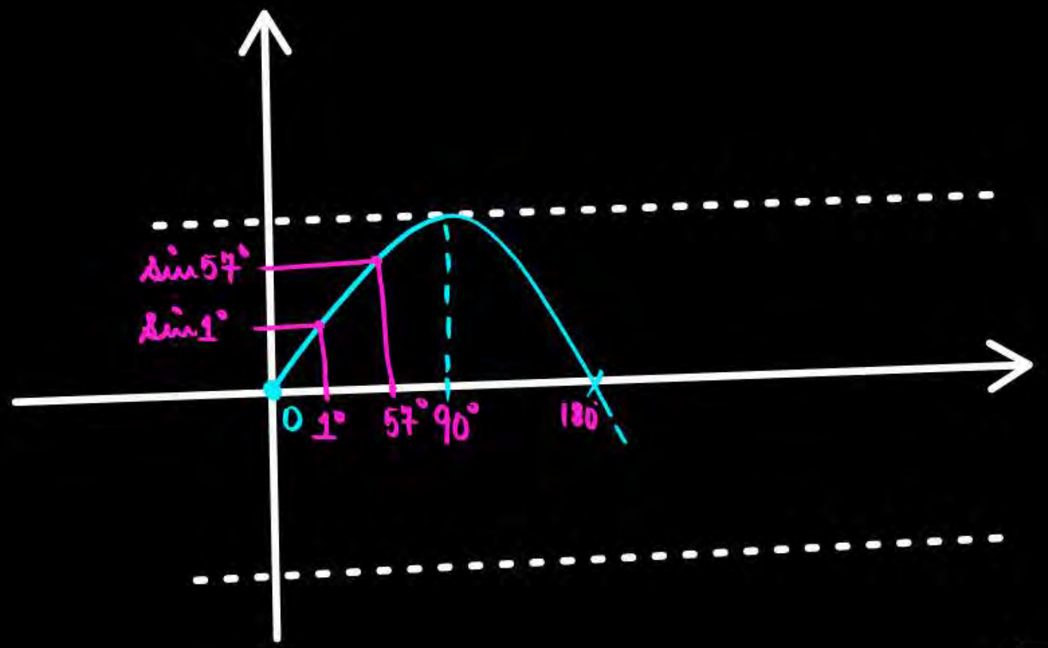
(ii) For $x \in [0, 2\pi]$: $\sin x$ decreases in $x \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ ✓

(iii) Arrange in increasing order : $\sin 1, \sin 2, \sin 3$.

(iv) Compare: $\sin 1^\circ$ & $\sin 1$

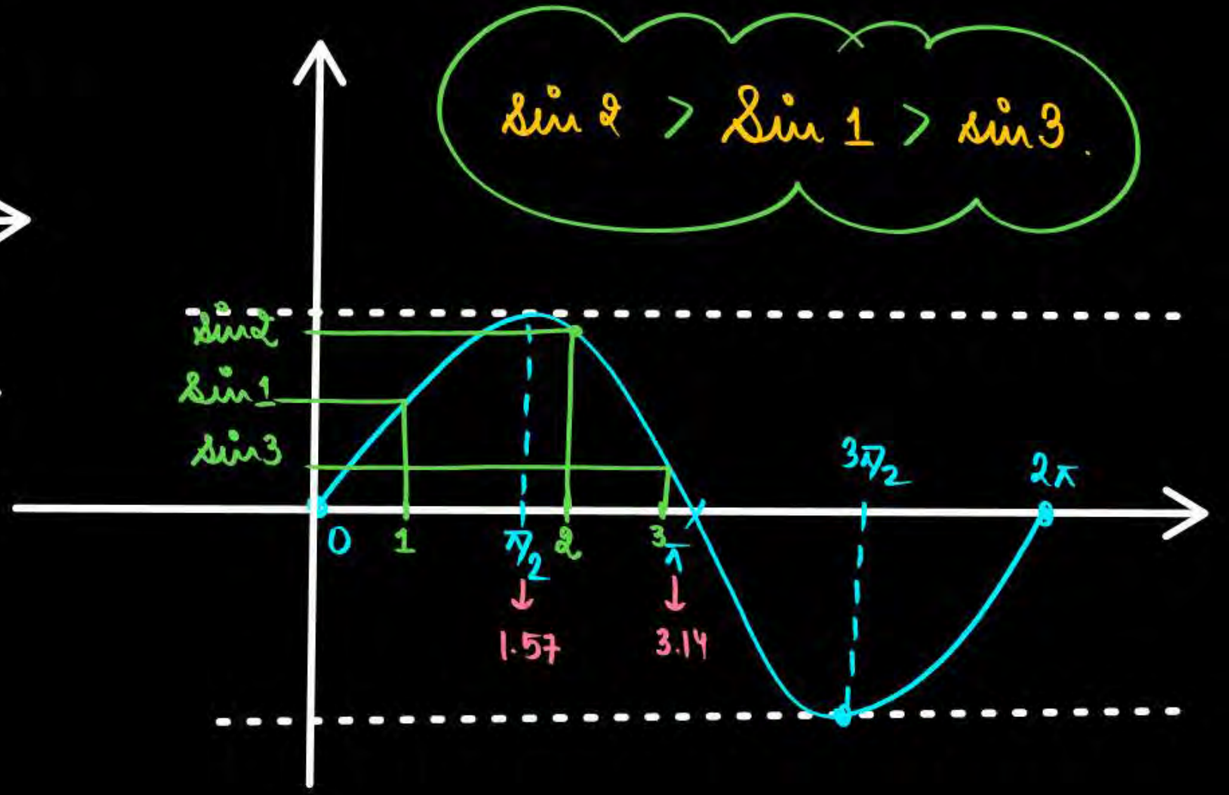
Radian





$\sin(1) \geq \sin 1^\circ$

$\simeq 57^\circ$



GRAPHS OF TRIGONOMETRIC FUNCTIONS

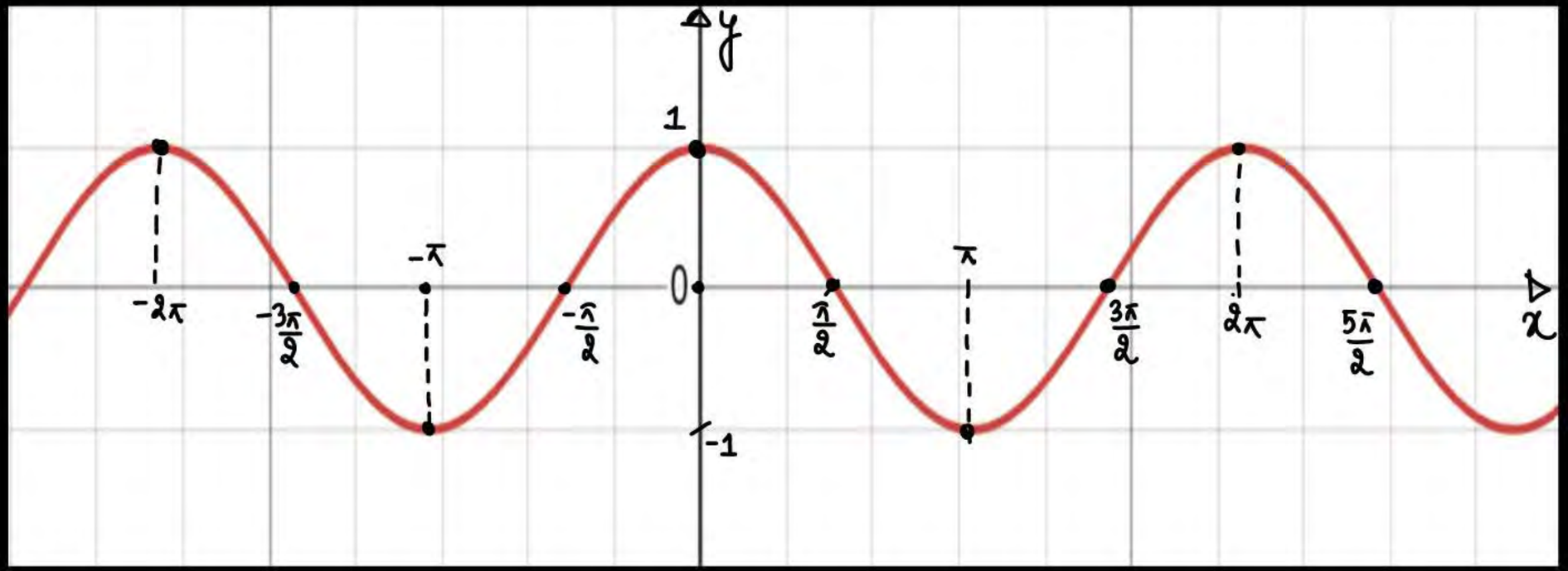
2. $y = \cos(x)$

domain $(x) \in R$

Range $(y) \in [-1, 1]$

GRAPHS OF TRIGONOMETRIC FUNCTIONS

2. $y = \cos(x)$



Period $= 2\pi$.

RAPID FIRE ROUND



Application of graph:

(i) For $x \in [0, 2\pi]$: $\cos x$ increases in _____.

(ii) For $x \in [0, 2\pi]$: $\cos x$ decreases in _____.

(iii) Arrange in increasing order : $\cos 1, \cos 2, \cos 3$.

(iv) Compare: $\cos 3$ & $\cos 4$



GRAPHS OF TRIGONOMETRIC FUNCTIONS

3. $y = \tan(x)$

$$y = \tan x = \frac{\sin x}{\cos x}$$

domain $\Rightarrow$ $\cos x \neq 0$

$$\text{domain}(x) \in R - \left\{(2n+1)\frac{\pi}{2}\right\}$$

$$\text{Range}(y) \in R$$

\# $\cos x = 0$

$$x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \pm\frac{7\pi}{2}, \pm\frac{9\pi}{2} \ldots$$

OR

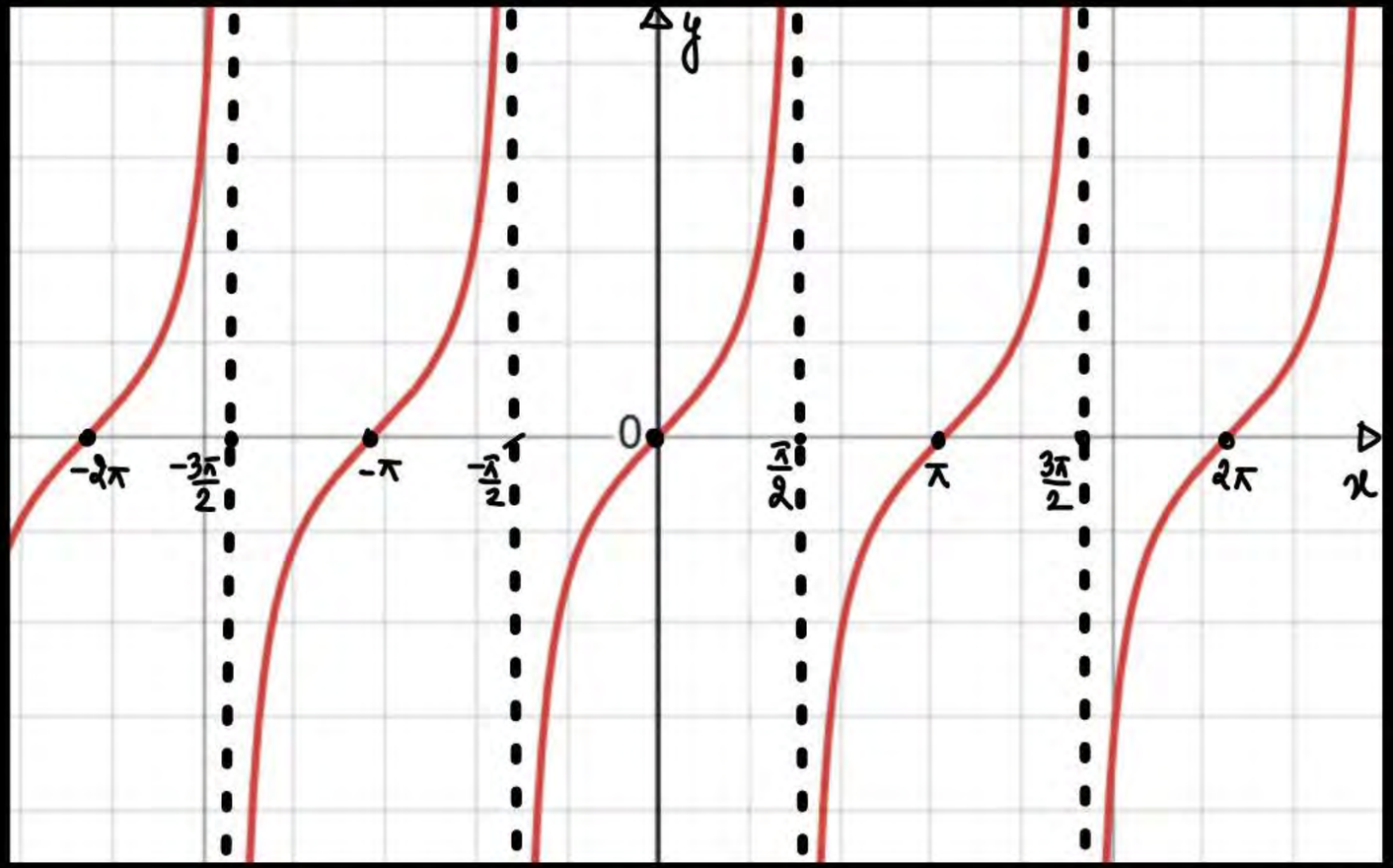
x = odd multiples of $\frac{\pi}{2}$.

OR

$$x = (2n+1)\frac{\pi}{2}, \quad n \in I.$$

GRAPHS OF TRIGONOMETRIC FUNCTIONS

3. $y = \tan(x)$



Period $= \pi$

GRAPHS OF TRIGONOMETRIC FUNCTIONS

4. $y = \cot(x)$

$y = \frac{\cos x}{\sin x}$

$\sin x \neq 0$

$\text{domain}(x) \in R - \{n\pi\}$

$\text{Range}(y) \in R$

$\sin x = 0$

$x = 0, \pi, 2\pi, 3\pi, 4\pi, \ldots$

$-\pi, -2\pi, -3\pi \ldots$

OR

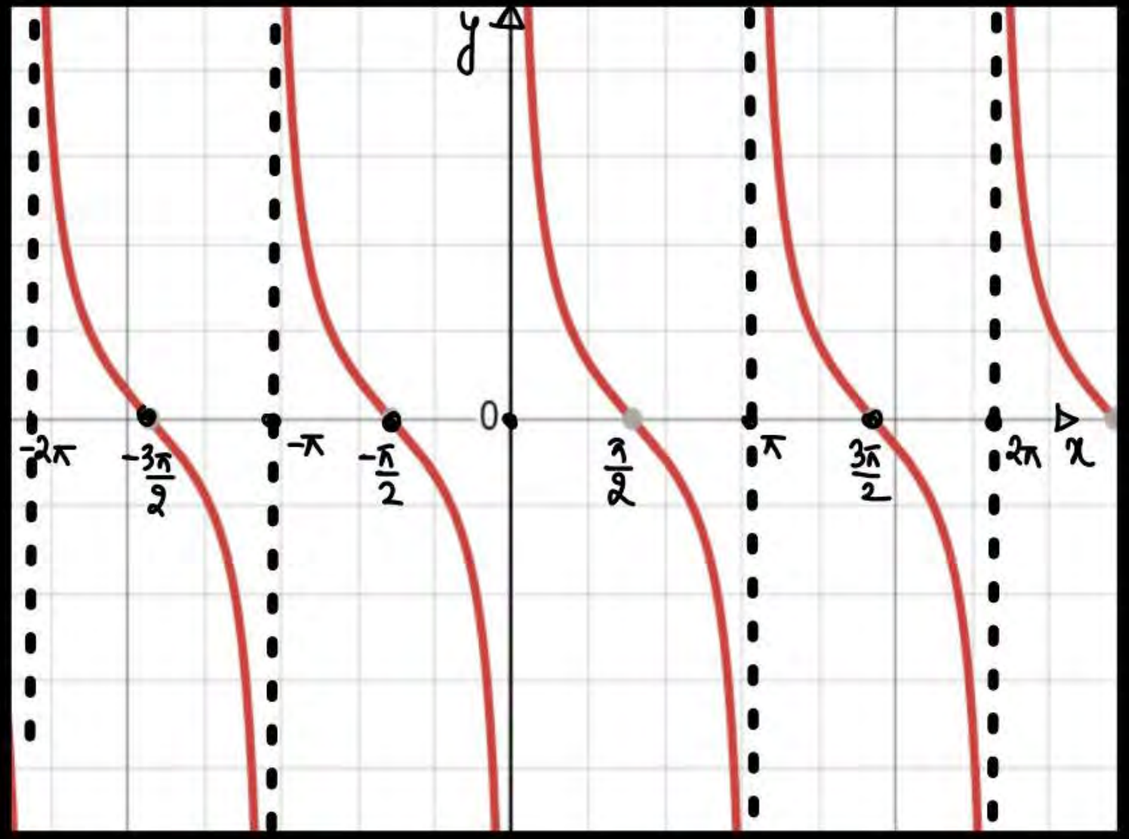
$x =$ every multiple of π.

OR

$x = n\pi, n \in I$.

GRAPHS OF TRIGONOMETRIC FUNCTIONS

4. $y = \cot(x)$



Period = π.



GRAPHS OF TRIGONOMETRIC FUNCTIONS

5. $y = \sec(x)$

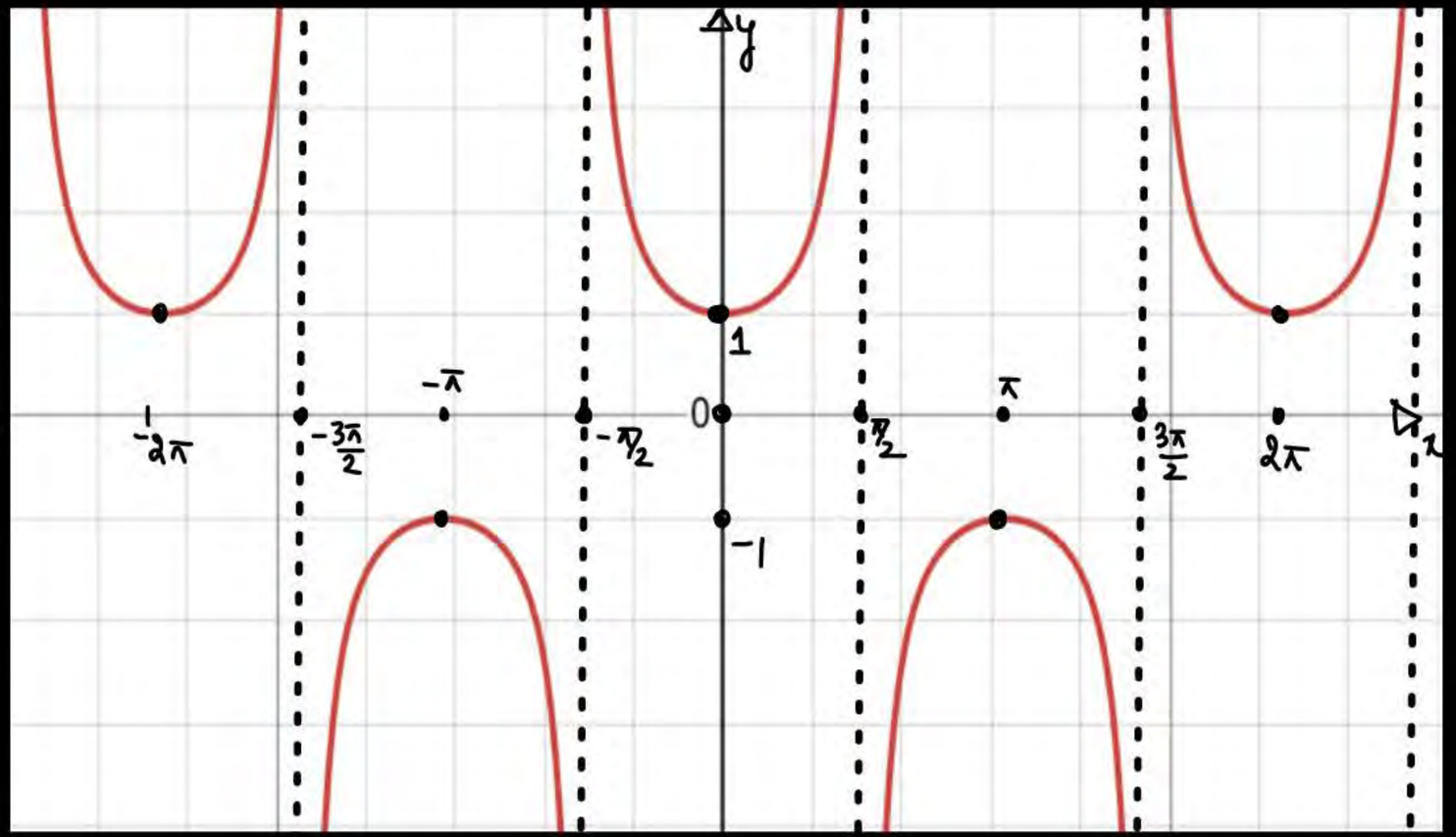
$$y = \sec(x) = \frac{1}{\cos x}$$

$$\text{domain}(x) \in R - \left\{(2n+1)\frac{\pi}{2}\right\}$$

$$\text{Range}(y) \in (-\infty, -1] \cup [1, \infty)$$

GRAPHS OF TRIGONOMETRIC FUNCTIONS

5. $y = \sec(x)$



Period $= 2\pi$

6. $y = \text{cosec}(x)$

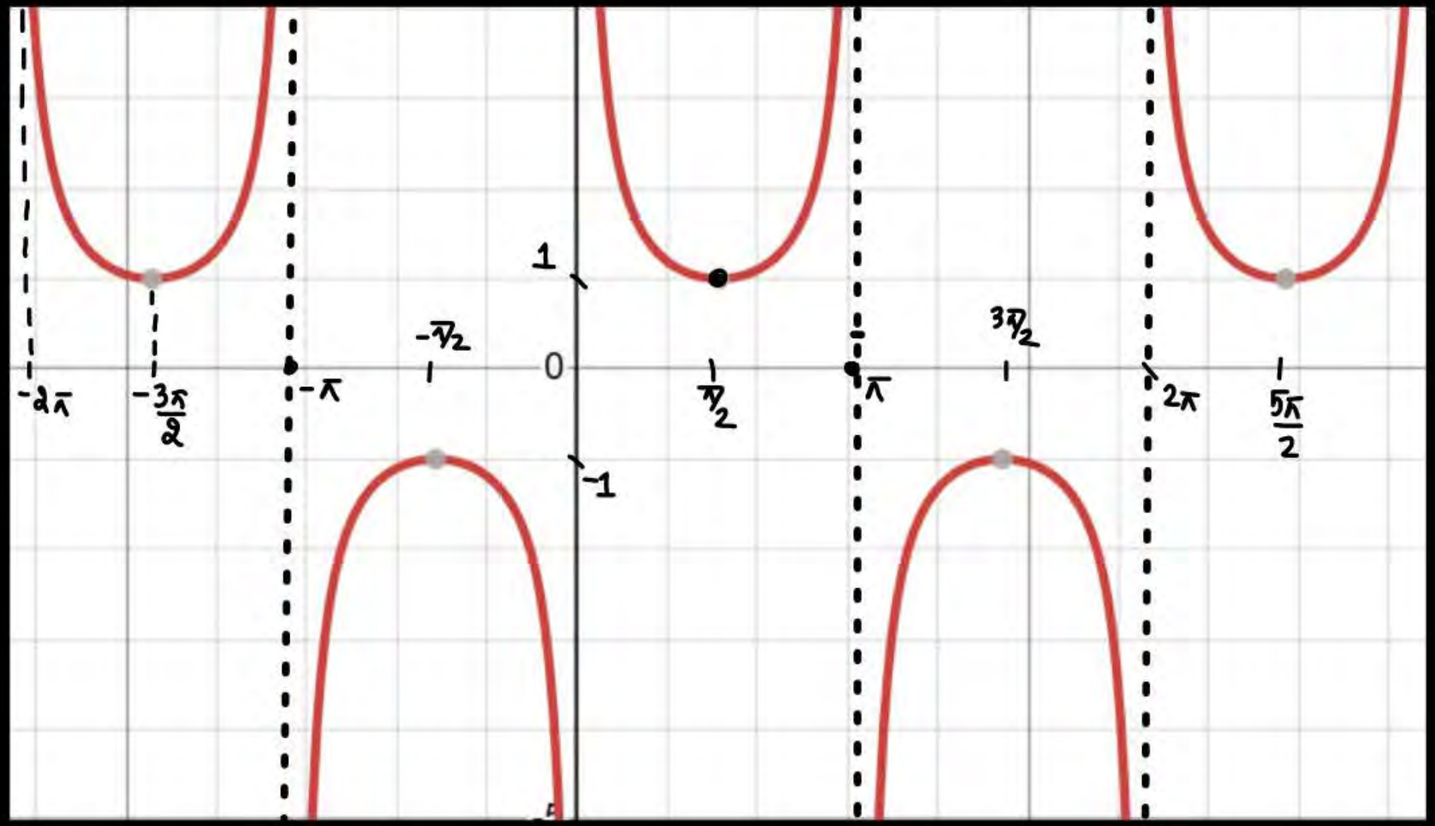
$y = \text{cosec}\,x = \frac{1}{\sin x}$

domain $(x) \in R - \{n\pi\}$

Range $(y) \in (-\infty, -1] \cup [1, \infty)$

GRAPHS OF TRIGONOMETRIC FUNCTIONS

6. $y = \text{cosec}(x)$



Period $= 2\pi$



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians